

TrES-3b

R-0063

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250515 · created 2026-07-15T15:15:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460810.8847 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.016
Transit depth (Rp/Rs)^2	2.73 +/- 0.52 [%]
Orbital Inclination (inc)	82.64 +/- 0.53
Airmass coefficient 1 (a1)	0.953 +/- 0.01
Airmass coefficient 2 (a2)	0.047 +/- 0.053
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	#2 - [292, 69]
Optimal Aperture	3.21
Optimal Annulus	12.43
Transit Duration (day)	0.0615 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.165 $\pm$ 0.016 vs 0.16309 (deviation 0.12 σ)
Duration fit vs published	0.0615 d vs 0.0567 d (deviation 1.07 σ)
Mid-time O-C	-5.6 min
Depth SNR	10.3
Residual scatter	1.09 %

Light curve

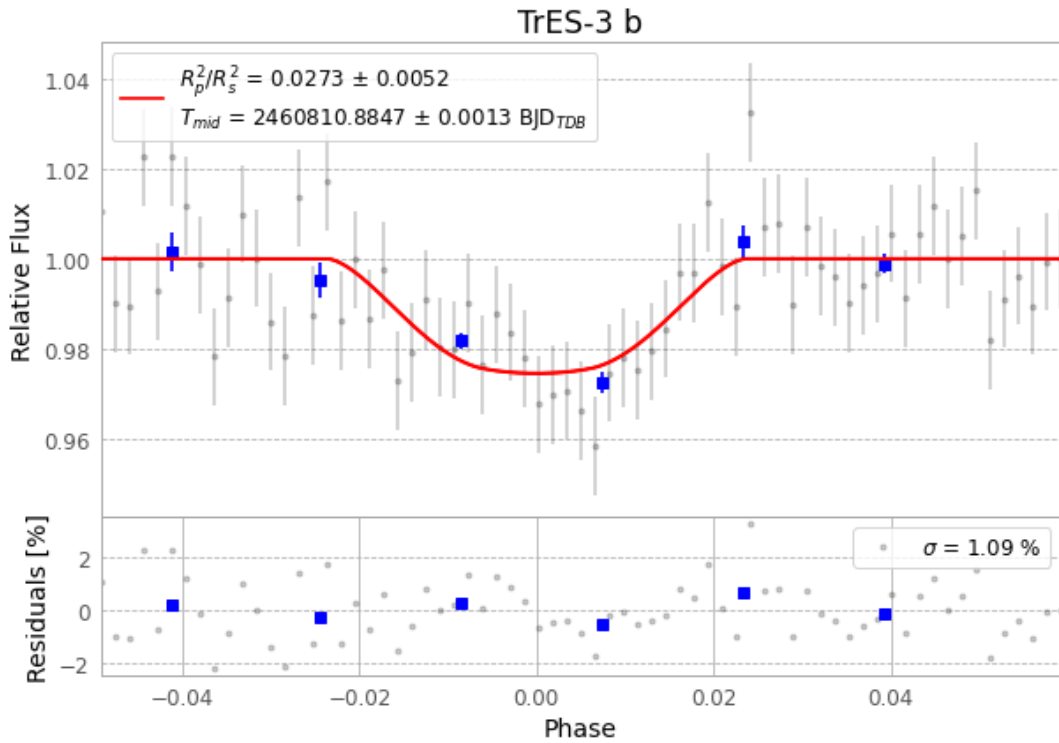


Figure 1 — FinalLightCurve_TrES-3 b_2025-05-15.png (SHA-256 8832525e5097ef65...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 229], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2e93f06cf1702e1f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

69 FITS frames (45 MB), TRES-3250515073616.FITS → TRES-3250515110016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250515.log (86979432f176...), FinalLightCurve_TrES-3 b_2025-05-15.png (8832525e5097...), FinalLightCurve_TrES-3 b_2025-05-15.csv (ecc032122f98...)

Compute resources

Wall time 385.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.